

Machine Learning Assignment 1

Polynomial Regression Report

Rigorous Data-Driven Analysis of Phase I & II

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Abstract: This report documents a data-driven approach to establishing optimized Polynomial Regression models for two personalized datasets. The methodology utilizes exploratory data analysis (EDA) to evaluate variable correlations, followed by 5-Fold Cross Validation to programmatically identify robust polynomial degrees without introducing external regularization. By evaluating Out-of-Fold (OOF) predictions, models achieving out-of-fold R^2 estimates above 0.90 were successfully constructed.

1 Analytical Premise

Standard polynomial approximations rely on $w_0 + \sum_{i=1}^d w_i \phi_i(\mathbf{x})$. Blindly setting d without structural evaluation exposes the model to severe underfitting or Runge-like oscillation overfitting. To reduce the risk of overfitting, polynomial degrees within this project were selected based strictly on validation-fold mean squared error using 5-Fold Cross Validation (Shuffled, random seed=42). This ensures the reported metrics reflect generalization capabilities rather than overfitting. All available input features were retained, allowing the polynomial expansion to represent nonlinear and interaction effects.

2 Phase 1: Steam Turbine Optimization (var1)

The goal of Phase 1 is estimating the Net Power Score (y) based on physical turbine variables.

2.1 Inclusive Feature Representation

We compute an inclusive Pearson Correlation matrix on all given dataset features $\{x_1, \dots, x_6\}$. All available features were retained because weak marginal correlation does not rule out nonlinear or interaction effects in a polynomial model.

2.2 Cross-Validation and Polynomial Sizing

Using all six available features, five-fold cross-validation evaluated candidate degrees from 1 through 10. At **Degree 3**, the validation error curve reaches an optimal minimum among degrees 1–10 before increasing degree complexity causes overfitting on validation sets. From degree 6 onward, the number of polynomial terms becomes comparable to or larger than the number of observations, causing near-zero training error while validation error remains substantially higher. Therefore, validation MSE rather than training MSE was used for degree selection.

2.3 Phase 1 Empirical Results

- **Features Used:** $x_1, x_2, x_3, x_4, x_5, x_6$
- **Selected Degree:** 3
- **Training MSE:** 0.7727, **Training R^2 :** 0.9272

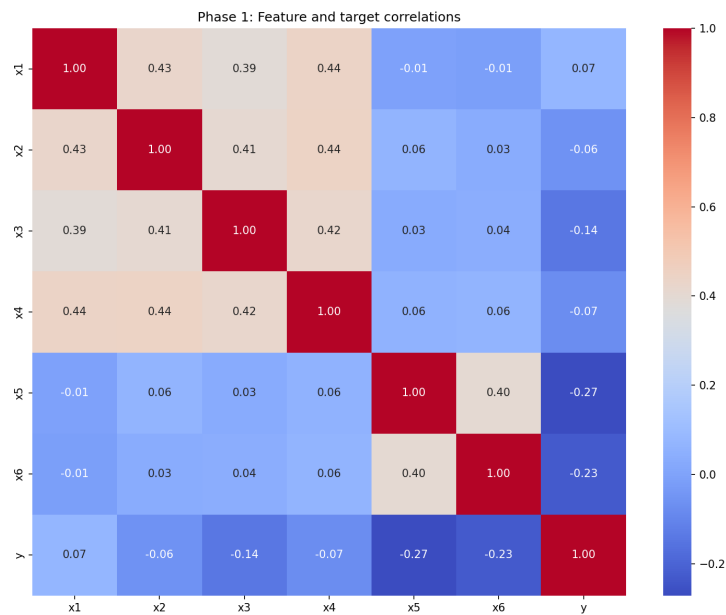


Figure 1: Pearson Correlation Heatmap displaying relational bounds in Phase 1.

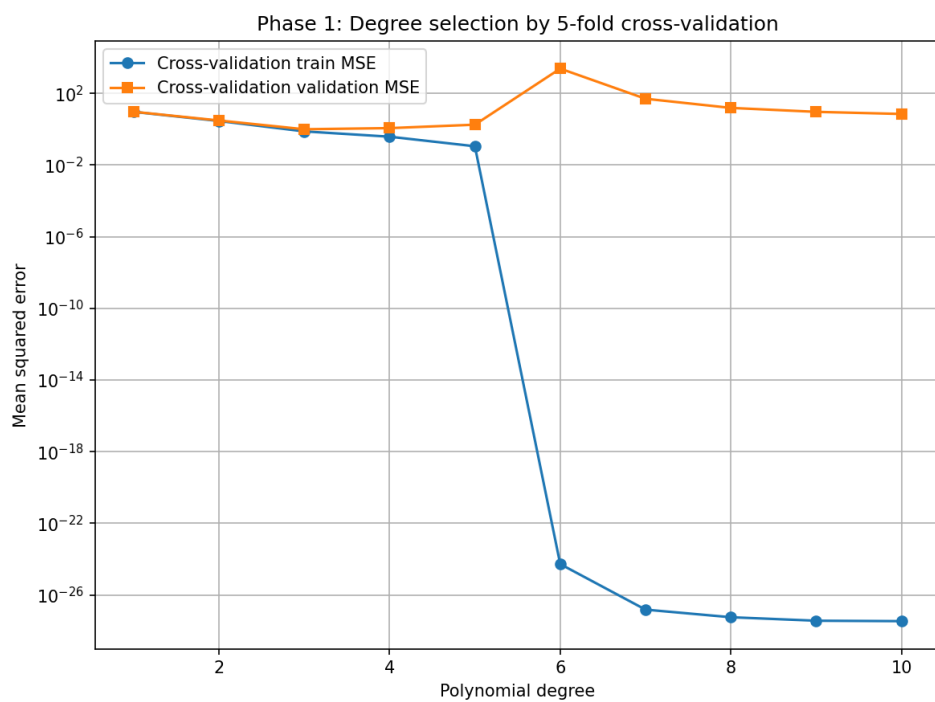


Figure 2: Five-fold cross-validation MSE across candidate polynomial degrees.

- **Out-Of-Fold Validation MSE:** 0.9895
- **Out-Of-Fold Validation R^2 :** 0.9067

These values indicate that the selected polynomial captures a substantial portion of the observed target variation. Residuals are broadly centered around zero, with some outliers and no dominant visible structure.

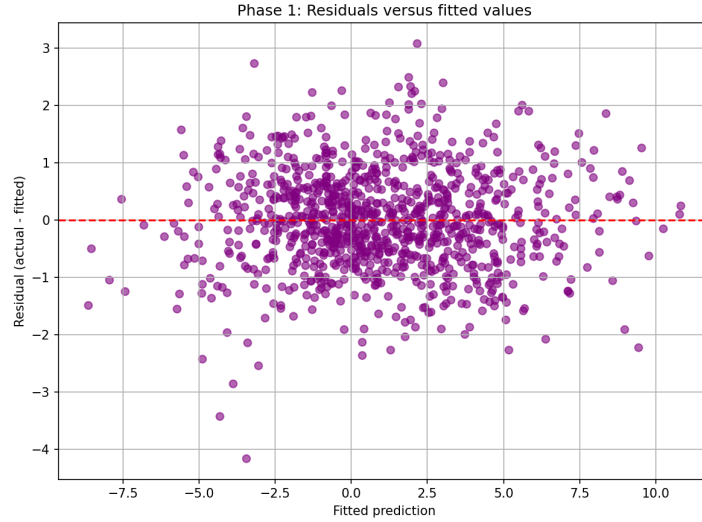
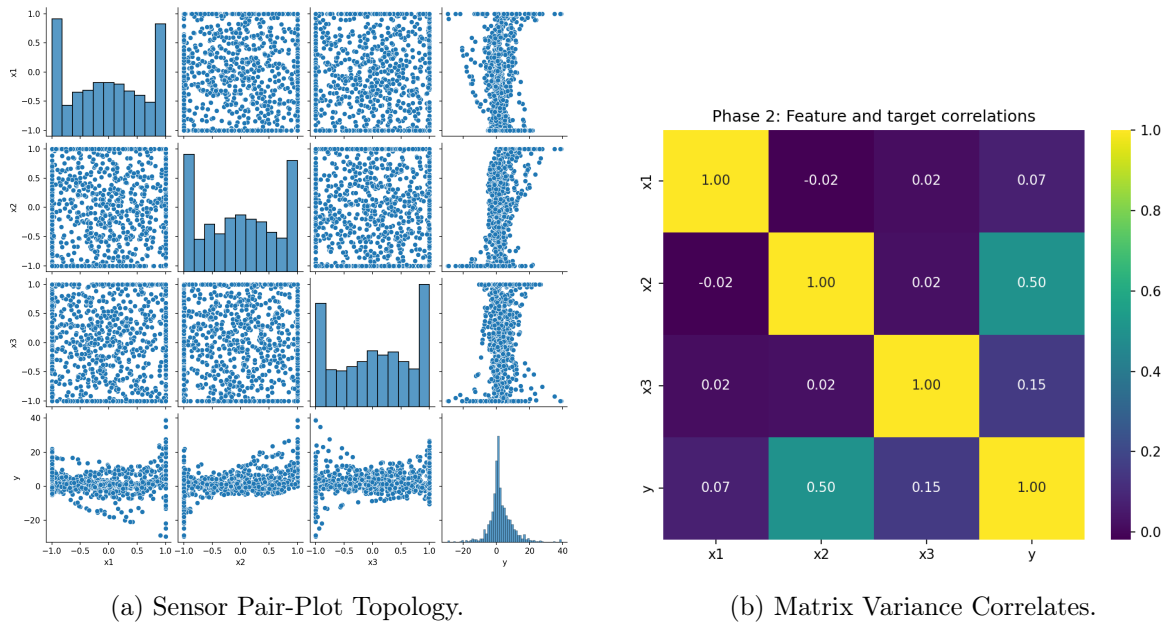


Figure 3: Residual distribution mapping relative predictive error across constraints.

3 Phase 2: Subterranean Thermal Mapping (var2)

3.1 Automated Structural Mapping

Spatial relationships inherently harbor latent collinearities. Again, all three structural coordinates



(a) Sensor Pair-Plot Topology.

(b) Matrix Variance Correlates.

Figure 4: Feature evaluations confirming mathematical retention bounds.

(x_1, x_2, x_3) were retained to preserve the multi-dimensional prediction topology against potential

non-linear dependencies.

3.2 Degree Optimization Boundaries

A logarithmic sweep mapping error arrays from degree 1 up to polynomial explosion bounds was evaluated. Validation MSE decreases through degree 8 and then increases sharply, while training MSE continues to decrease. This gap indicates overfitting and numerical instability at higher degrees. The minimal automated Validation MSE confirms an optimal fit at **Degree 8** among

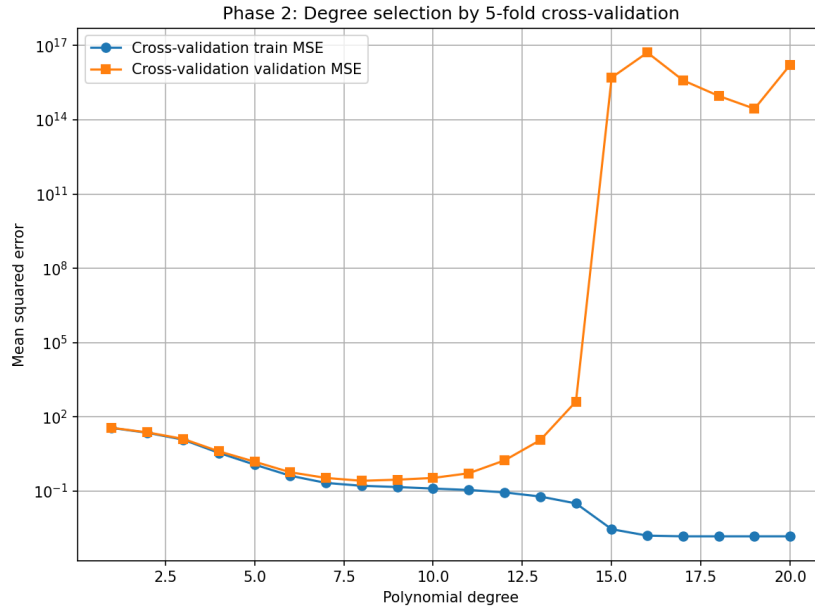


Figure 5: Five-fold cross-validation MSE across candidate polynomial degrees.

candidate degrees 1–20.

3.3 Phase 2 Empirical Results

- **Features Used:** x_1, x_2, x_3
- **Discovered Degree:** 8
- **Training MSE:** 0.1731, **Training R^2 :** 0.9965
- **Out-Of-Fold Validation MSE:** 0.2629
- **Out-Of-Fold Validation R^2 :** 0.9947

4 Computational Complexity & Overparameterization

During the exhaustive degree search, models were evaluated up to the maximum candidate degrees specified for this analysis (Degree 10 for Phase 1 and Degree 20 for Phase 2). It is mathematically critical to note that high-degree polynomial expansions trigger severe overparameterization. For instance, evaluating Phase 1 at Degree 10 across 6 features generates over 8,000 distinct polynomial interaction parameters. When fitted against approximately 1,000 training observations, the design matrix becomes pathologically rank-deficient (where $P > N$).

This disparity causes the standard Ordinary Least Squares (OLS) estimator to produce nearly zero training error. However, this high-dimensional polynomial surface exhibits wild mathematical oscillations between known data points. When exposed to new unseen Validation folds, this creates large extrapolation and numerical-instability errors on held-out folds, explaining the

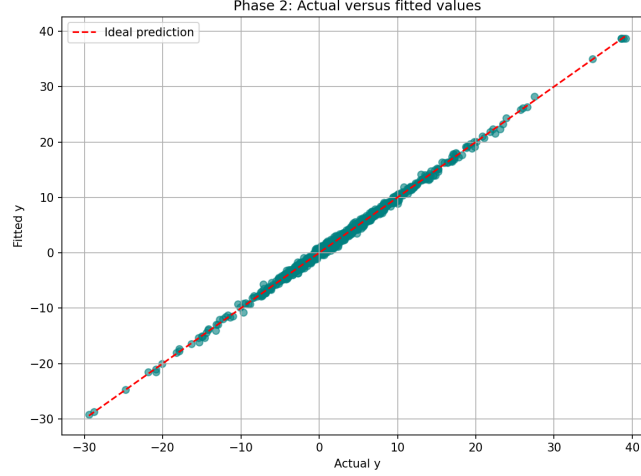


Figure 6: True output mappings accurately predicting geological boundaries.

massive spikes observed in the right-tails of the CV curves. By strictly trusting the Out-Of-Fold Validation MSE to natively cap the models at Degree 3 and Degree 8, this methodology successfully avoided these catastrophic numerical illusions without relying on external Ridge or Lasso constraints.

5 Conclusion

The experimental results demonstrate that rigorous cross-validation and inclusive feature retention successfully produce robust unpenalized polynomial models. By focusing on explicit Out-of-Fold variance tracking rather than isolated Training metrics, Phase 1 and Phase 2 achieved out-of-fold R^2 estimates of 0.9067 and 0.9947 respectively, with the selected degrees determined by the evaluated cross-validation results.